A NOVEL LOG-NORMAL DISTRIBUTION: SIMULATION, APPLICATIONS TO MECHANICAL COMPONENT DATA AND ENVIRONMENTAL DATA

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Abstract

The normal distribution and its exponentiated counterpart the log-normal distribution are widely recognized as fundamental tools in statistical analysis. In this study, we propose a new extension of the log-normal family and undertake a detailed examination of its structural characteristics, including theoretical properties, simulation behaviour, and practical applicability. In this paper, we derive the expressions and examines the three-parameter Weighted log-normal distribution. Probability functions, survival and hazard structures, quantiles function and moment properties. We estimates the parameters through maximum likelihood estimation and supported through extensive Monte Carlo investigations designed to assess the estimator's stability, systematic deviation, and overall inferential performance. he estimation of parameters is performed with the aid of the L-BFGS-B optimization algorithm in Python `scipy.optimize.minimize` library, ensuring precise computational results. In empirical analyses involving diverse real-world datasets, the proposed model consistently demonstrates superior adequacy across all three datasets derived from mechanical component failures and environmental measurements.

Keywords. Weighted distribution, Maximum likelihood estimation, Simulation, Mechanical data, Environmental Data.

Mathematics Subject Classifcation. 62G30, 62E10, 62E15, 62F10.

1 Introduction

In recent decades, researchers have expanded and modified statistical models to handle real-world data from various fields. Traditional probability distributions often fail to adequately describe some forms of data, which motivates the development of new families that either build upon existing models or create hybrids with additional characteristics for greater flexibility. The normal distribution, and log-normal distribution [15] in particular, have attracted a lot of attention due to their many applications and inherent flexibility. It continues to serve as the foundation for many of the generalized variants proposed in the literature because of these advantages. The use of statistical models enhances informed judgment and plays a vital role in expanding theoretical and empirical understanding across multiple fields of study, see [8], [9], [10], [11], [13], [14], [18], [19], [21], [23], [26], [27], [29], [31].

Let X be a non-negative random variable with the PDF $g(x)$ and suppose non-negative weight function is $w(x)$, then the PDF of the weighted random variable X_w is given by

$$g_w(x) = \frac{w(x)\, f(x)}{E(w(x))}, \quad x > 0, \tag{1}$$

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where $w(x)$ be a non-negative weight function and

$$E(w(x)) = \int_{-\infty}^{+\infty} w(x)g(x)dx < \infty.$$

Let $w(x)$ denote a non-negative weighting function for which the expectation $E[w(x)]$ is finite. Rao [24] introduced a particular form of weighted distributions, referred to as size biased distributions, by selecting $w(x) = x^c$. This formulation is known as the size-biased distribution of order c. Specifically, when $c = 1$, it is identified as the length biased distribution, and when $c = 2$, it is termed the area-biased distribution. For more deatails [7], [12], [17], [22], [25], [28].

The probability density function associated with the random variable X in this distribution is expressed as,

$$f_{LN}(x) = \frac{1}{x\,\sigma\sqrt{2\pi}}\, exp\left[-\frac{1}{2}\left(\frac{\ln x - \mu}{\sigma}\right)^2\right] \qquad -\infty < \mu < \infty,\ \sigma > 0. \tag{2}$$

and the cumulative distribution function (*cdf*) is

$$F_{LN}(x) = \frac{1}{2}\left[1 + erf\left(\frac{\log x - \mu}{\sigma\sqrt{2}}\right)\right] = \Phi\left(\frac{\log x - \mu}{\sigma\sqrt{2}}\right) \tag{3}$$

where

$$erf(z) = \frac{2}{\pi}\int_0^z e^{t^2}dt$$

and

$$\Phi(z) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^z e^{-t^2/2}dt$$

From (1) and (2), we get the *pdf* and *cdf* of weighted log-normal distribution (WLN) is given as

$$f_{WLN}(x) = \frac{1}{x\,\sigma\sqrt{2\pi}}\, exp\left[-\frac{1}{2}\left(\frac{\ln x - (\mu + c\sigma^2)}{\sigma}\right)^2\right] \qquad -\infty < \mu < \infty,\ \sigma > 0 \text{ and } c > 0. \tag{4}$$

and the cumulative distribution function (*cdf*) is

$$F_{WLN}(x) = \Phi\left[\frac{\ln x - (\mu + c\sigma^2)}{\sigma}\right] \qquad -\infty < \mu < \infty,\ \sigma > 0 \text{ and } c > 0. \tag{5}$$

Now, $c = 0$ and $c = 1$ find the log-normal distribution and length biased log-normal distribution respectively. The structure of this document is as follows. The Weighted log-normal distribution's probability density function and cumulative distribution function are derived in Section 1. Its mathematical characteristics, such as survival and hazard functions, quantile function, and moments, are then discussed in Section 2. Section 3 presents the maximum likelihood (ML) approaches for parameter estimation. The results of a simulation study that assesses the effectiveness of the suggested estimators are presented in Section 4, and applications to real-world datasets in Section 5 demonstrate the model's usefulness. Section 6 provides a summary of the concluding thoughts.

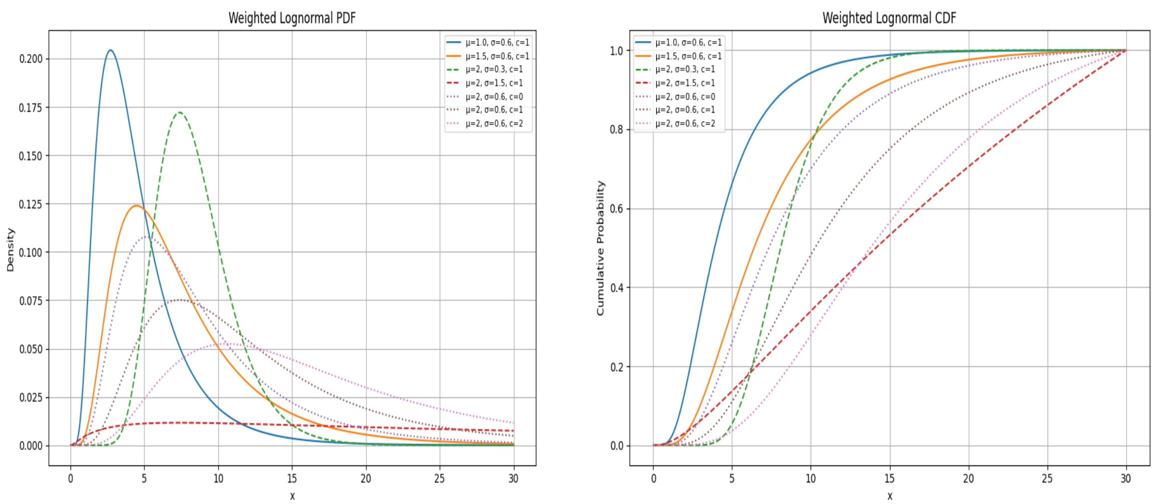


Figure 1: Behaviour of the weighted log-normal distribution at different values of different parameters

2 Statistical Properties

In this section, we proceed to establish closed-form expressions for essential probabilistic measures of the proposed distribution, namely the survival and hazard functions, the quantile function, the sequence of raw moments, Shannon's entropy, and the mean residual life function.

2.1 Survival and Hazard functions

The survival function characterizes lifetime distribution and is essential in modelling time-to-event data. It quantifies the event probability over time and its functional form is given as

$$S(x) = P(X > x) = \int_x^\infty f(x)dx = 1 - \Phi\left[\frac{\ln x - (\mu + c\sigma^2)}{\sigma}\right] \qquad -\infty < \mu < \infty,\ \sigma > 0 \text{ and } c > 0. \tag{6}$$

The hazard function is given by

$$h(x) = \frac{f(s)}{S(x)} = \frac{1}{x\,\sigma\sqrt{2\pi}}\, exp\left[-\frac{1}{2}\left(\frac{\ln x - (\mu + c\sigma^2)}{\sigma}\right)^2\right] \div \left\{1 - \Phi\left[\frac{\ln x - (\mu + c\sigma^2)}{\sigma}\right]\right\} \tag{7}$$

Quantile Function

The quantile function, denoted by $Q(p)$, gives the value x_p corresponding to a specific cumulative probability p, where $0 < p < 1$. It is defined as the inverse of the cumulative distribution function as

$$Q(p) = F^{-1}(p) = \inf\{x \in \mathbb{R} : p \leq F(x)\}, \qquad 0 < p < 1. \tag{8}$$

For the weighted log-normal distribution, the *cdf* is given by

$$F(x;\mu,\sigma,c) = \Phi\left(\frac{\ln x - (\mu + c\sigma^2)}{\sigma}\right),$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. To derive the quantile function $Q(p)$, we solve $F(x_p) = p$ for x_p as,

$$p = \Phi\left(\frac{\ln x_p - (\mu + c\sigma^2)}{\sigma}\right).$$

The inverse of the standard normal *cdf*, we obtain

$$\Phi^{-1}(p) = \frac{\ln x_p - (\mu + c\sigma^2)}{\sigma}.$$

Rearranging for x_p, the quantile function becomes:

$$Q(p) = x_p = \exp\left[\mu + c\sigma^2 + \sigma\Phi^{-1}(p)\right]. \tag{9}$$

For $p = 0.5$, we obtain the median x_m. Since $\Phi^{-1}(0.5) = 0$, the median is

$$x_m = \exp(\mu + c\sigma^2).$$

Table-1, explain the the behaviour of WLN distribution based on quantile values at the different values of different parameters values. Median are given at 50% in the table values.

Table 1: The weighted Log-normal quantiles ($\mu = 0.5$, $\sigma = 0.8$)

Percentile	**c = -1.0**	**c = 0.0**	**c = 0.5**	**c = 1.0**
1%	0.135189	0.256384	0.353074	0.486228
10%	0.311847	0.591413	0.814451	1.121603
25%	0.506824	0.961182	1.323670	1.822863
50%	0.869358	1.648721	2.270500	3.126768
75%	1.491216	2.828062	3.894603	5.363366
90%	2.423569	4.596252	6.329627	8.716705
99%	5.590554	10.602378	14.600829	20.107207

2.2 Moments

Moments are essential instruments for characterising a probability distribution's structure. They offer a consistent framework for describing important statistical characteristics, such as skewness, kurtosis, variability, and central tendency. Moments are essentially numerical descriptors that highlight significant aspects of a distribution, [1], [2], [3], [4], [5], [6].
The k-th, order moment about the origin of (2) is given by

$$E[X^k] = \int_0^\infty x^k f_{\mathrm{WLN}}(x)\,dx = \exp\left(k(\mu + c\sigma^2) + \frac{1}{2}k^2\sigma^2\right). \tag{10}$$

For $k = 1, 2, 3, 4$, the first four raw moments are

$$E[X] = \exp\left(\mu + c\sigma^2 + \frac{1}{2}\sigma^2\right),$$

$$E[X^2] = \exp\left(2\mu + 2c\sigma^2 + 2\sigma^2\right),$$

$$E[X^3] = \exp\left(3\mu + 3c\sigma^2 + \frac{9}{2}\sigma^2\right)$$

and

$$E[X^4] = \exp\left(4\mu + 4c\sigma^2 + 8\sigma^2\right).$$

The variance is given as

$$V(X) = E[X^2] - \left(E[X]\right)^2 = \exp\left(2\mu + 2c\sigma^2 + \sigma^2\right)\left(\exp(\sigma^2) - 1\right). \tag{11}$$

The skewness and excess kurtosis can be obtained as

$$\gamma_1 = \left(\exp(\sigma^2) + 2\right)\sqrt{\exp(\sigma^2) - 1} \tag{12}$$

$$\gamma_2 = \exp(4\sigma^2) + 2\exp(3\sigma^2) + 3\exp(2\sigma^2) - 6, \tag{13}$$

respectively.

Table 2: Weighted Lognormal Moments, Variance, Skewness and Kurtosis

σ	$E(X)$	$E(X)^2$	$E(X)^3$	$E(X)^4$	$V(X)$	Skewness	Kurtosis
1.0	0.886323	1.227040	2.434382	6.518177	0.441470	1.923642	9.337117
1.5	0.632732	0.798335	1.595075	4.402710	0.397985	2.335206	11.380186
2.0	0.476833	0.561347	1.124345	3.225253	0.333977	2.788348	15.164502
2.5	0.381562	0.429636	0.859632	2.493566	0.284047	3.163684	18.507937
3.0	0.314491	0.340033	0.659942	1.830150	0.241129	3.389526	20.164180
3.5	0.266567	0.286425	0.576870	1.717812	0.215367	3.859043	26.080294
4.0	0.230828	0.243237	0.478129	1.366812	0.190207	4.032481	27.511965
4.5	0.204058	0.213913	0.432950	1.339213	0.172273	4.461228	34.842986
5.0	0.183100	0.191495	0.377461	1.057027	0.157969	4.532114	32.688790
5.5	0.162414	0.164333	0.316582	0.880086	0.137955	4.783054	36.693829

3 Estimation

Using observed data, parameter estimate is the process of identifying numerical values that define a statistical model. Models to describe the relationships between variables are commonly developed in fields including physics, biology, engineering, and economics. Parameters

Table 3: Weighted Lognormal Moments, Variance, Skewness and Kurtosis

μ	$E(X)$	$E(X^2)$	$E(X^3)$	$E(X^4)$	$V(X)$	Skewness	Kurtosis
1.0	1.118320	1.870416	4.332974	13.177047	0.619777	1.752388	8.167798
1.5	1.386138	2.753844	7.338967	25.008354	0.832466	1.598253	7.199212
2.0	1.665163	3.839550	11.604984	44.283736	1.066783	1.505474	6.853394
2.5	1.970213	5.203717	17.560535	72.891612	1.321978	1.380912	6.004014
3.0	2.300450	6.978311	26.669922	124.613703	1.686243	1.305414	5.895271
3.5	2.696413	9.232536	38.771967	194.104037	1.961893	1.199825	5.220855
4.0	2.952358	10.847129	48.278808	254.678553	2.130712	1.180941	5.263095
4.5	3.417621	14.356789	72.403638	427.165037	2.676655	1.151365	4.777872
5.0	3.588316	16.202316	88.537931	565.947455	3.326304	1.075937	4.473070
5.5	3.891197	18.057597	100.336815	654.823707	2.916185	1.481198	5.388039

Table 4: Weighted Lognormal Moments, Variance, Skewness and Kurtosis

c	$E(x)$	$E(x)^2$	$E(x)^3$	$E(x)^4$	$V(x)$	Skewness	Kurtosis
1.0	0.880436	1.213517	2.385235	6.251802	0.438349	1.877652	8.811007
1.5	0.720257	0.778767	1.167087	2.297010	0.259997	1.747319	8.154609
2.0	0.608947	0.545906	0.668390	1.055972	0.175900	1.675039	7.502063
2.5	0.536865	0.414931	0.434166	0.588002	0.126707	1.670737	7.722850
3.0	0.480104	0.327628	0.296638	0.340802	0.096768	1.547972	6.872591
3.5	0.436221	0.268263	0.218315	0.223395	0.077974	1.527788	6.598015
4.0	0.400362	0.223649	0.164706	0.152926	0.063359	1.531923	6.768692
4.5	0.374692	0.193136	0.138092	0.108412	0.052742	1.452672	6.435146
5.0	0.347201	0.166743	0.104642	0.082386	0.046194	1.477725	6.591647
5.5	0.325831	0.145701	0.084514	0.061017	0.039536	1.434361	6.311864

are quantifiable measurements that control the structure and behaviour of these models. The process of estimating parameters entails determining which values best capture the data, guaranteeing that the model faithfully captures the fundamental connections between the variables being studied. In this section,we consider estimation of parameters using method of Maximum Likelihood Estimation.

3.1 Maximum Likelihood Estimation

Using the maximum likelihood approach to estimate the parameters of the WLN distribution is the main emphasis of this section. Consider $X_1, X_2, \ldots, X_n$ to be a random sample of size n from a population that follows the WLN distribution, and $x_1, x_2, \ldots, x_n$ to represent a random sample of size n taken from a population that follows the WLN distribution with observed values. For a random sample $x_1, x_2, \ldots, x_n$, the log-likelihood function for the WLN distribution is expressed as

$$\ell(\mu, \sigma, c) = \sum_{i=1}^{n} \ln f(x_i; \mu, \sigma, c).$$

Substituting the expression for $f(x_i; \mu, \sigma, c)$, we obtain

$$\ell(\mu, \sigma, c) = (c-1)\sum_{i=1}^{n} \ln x_i - n\ln(\sigma\sqrt{2\pi}) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(\ln x_i - \mu)^2 - n\left(c\mu + \frac{1}{2}c^2\sigma^2\right).$$

To obtain the MLEs of μ, σ, and c, the first partial derivatives of $\ell(\mu, \sigma, c)$ with respect to each parameter are taken and set to zero,

$$\frac{\partial \ell}{\partial \mu} = 0, \qquad \frac{\partial \ell}{\partial \sigma} = 0, \qquad \frac{\partial \ell}{\partial c} = 0.$$

These equations are nonlinear and are therefore solved numerically using computational methods the L-BFGS-B optimization algorithm in Python `scipy.optimize.minimize` library.

4 Simulation Study and Data Analysis

In order to evaluate the performance of maximum likelihood (ML) estimators, we conduct a simulation study based on bias and mean squared error (MSE). Independent random samples of sizes $n = 20, 30, 40, 50$ and 100 are generated with 1000 replications for each case. The parameter settings for ($\mu = 8.66418$, $\sigma = 0.57667$, $c = 1.019173$), ($\mu = -2.20529$, $\sigma = 0.415457$, $c = 0.98899$) and ($\mu = 3.958446$, $\sigma = 1.555012$) are chosen. The outcomes, summarized in Tables 5, 6 and 7 are indicate that both the Bias and the MSE progressively decline toward zero, consistent with the theoretical expectation of asymptotically unbiased and efficient estimators.

Algorithm: Simulation Procedure for the Parameter Estimation
Input: – Dataset D with N observations;
– Sample sizes $n \in \{20, 30, 40, 50, 100\}$;
– Number of replications $R = 1000$;
– The L-BFGS-B optimization algorithm in Python `scipy.optimize.minimize` library Function `mle_estimate()` returning $(\hat{\mu}, \hat{\sigma}, \hat{c})$.
For each dataset D_i, $i = 1, \dots, N$:

Step 1 (Compute true parameters):
Use the full dataset D_i to compute
$(\mu_{\text{true}}, \sigma_{\text{true}}, c_{\text{true}}) =$ `mle_estimate`(D_i).

Step 2 (Simulation study):
For each sample size n:
Initialize empty vectors: $\hat{\mu}, \hat{\sigma}, \hat{c}$.

For each replication $r = 1, \dots, R$:
Draw a bootstrap sample S_r of size n (with replacement).
Compute $(\mu_r, \sigma_r, c_r) =$ `mle_estimate`(S_r).
Store μ_r in $\hat{\mu}$, σ_r in $\hat{\sigma}$, c_r in $\hat{c}$.
End

Performance measures:
$\text{Bias}(\hat{\mu}) = \text{mean}(\hat{\mu}) - \mu_{\text{true}}$
$\text{Bias}(\hat{\sigma}) = \text{mean}(\hat{\sigma}) - \sigma_{\text{true}}$
$\text{Bias}(\hat{c}) = \text{mean}(\hat{c}) - c_{\text{true}}$

$\text{SE}(\hat{\mu}) = \text{sd}(\hat{\mu}), \quad \text{SE}(\hat{\sigma}) = \text{sd}(\hat{\sigma}), \quad \text{SE}(\hat{c}) = \text{sd}(\hat{c})$
$\text{MSE}(\hat{\mu}) = \text{mean}[(\hat{\mu} - \mu_{\text{true}})^2]$
$\text{MSE}(\hat{\sigma}) = \text{mean}[(\hat{\sigma} - \sigma_{\text{true}})^2]$
$\text{MSE}(\hat{c}) = \text{mean}[(\hat{c} - c_{\text{true}})^2]$

Figure 2: Simulation algorithm used for evaluating bias, standard error, and mean squared error of MLEs under the weighted log-normal distribution.

5 Application and Evaluation indicators through MLEs

This section presents an empirical study of the proposed WLN distribution using a real data set. The estimation of parameters is performed with the aid of the L-BFGS-B optimization algorithm in Python `scipy.optimize.minimize` library, ensuring precise computational results. Model adequacy is examined through several well-known information-based criteria, namely the Akaike Information Criterion (AIC), the Bayesian Information Criterion (BIC), the Corrected Akaike Information Criterion (AICC), and the Hannan–Quinn Information

Table 5: Simulation results for Weighted Log-normal Distribution $\mu = 8.66418,\ \sigma = 0.57667,\ c = 1.019173,$ True values-Data Set-I

n	$\hat{\mu}$	$\hat{\sigma}$	$\hat{c}$	SE($\hat{\mu}$)	SE($\hat{\sigma}$)	SE($\hat{c}$)	Bias($\hat{\mu}$)	Bias($\hat{\sigma}$)	Bias($\hat{c}$)	MSE($\hat{\mu}$)	MSE($\hat{\sigma}$)	MSE($\hat{c}$)
20	8.652529	0.553274	1.017935	0.145393	0.102611	0.007073	-0.011650	-0.023396	-0.001238	0.021254	0.011066	0.000052
30	8.659296	0.557323	1.018048	0.120612	0.079826	0.005596	-0.004884	-0.019347	-0.001125	0.014557	0.006740	0.000033
40	8.658956	0.564496	1.018445	0.109594	0.072453	0.005178	-0.005223	-0.012174	-0.000727	0.012026	0.005392	0.000027
50	8.655241	0.561580	1.018255	0.106885	0.073216	0.005284	-0.008939	-0.015090	-0.000918	0.011493	0.005583	0.000029
100	8.660196	0.565966	1.018578	0.104436	0.073089	0.005273	-0.003983	-0.010704	-0.000595	0.010912	0.005451	0.000028

Table 6: Simulation results for Weighted Log-normal Distribution $\mu = -2.20529,\ \sigma = 0.415457,\ c = 0.98899,$ True values-Data Set-II

n	$\hat{\mu}$	$\hat{\sigma}$	$\hat{c}$	SE($\hat{\mu}$)	SE($\hat{\sigma}$)	SE($\hat{c}$)	Bias($\hat{\mu}$)	Bias($\hat{\sigma}$)	Bias($\hat{c}$)	MSE($\hat{\mu}$)	MSE($\hat{\sigma}$)	MSE($\hat{c}$)
20	-2.205990	0.381773	0.990323	0.099884	0.124702	0.006468	-0.000700	-0.033684	0.001332	0.009967	0.016670	0.000044
30	-2.204427	0.384476	0.990187	0.106015	0.128671	0.006527	0.000863	-0.030981	0.001197	0.011229	0.017500	0.000044
40	-2.213666	0.379845	0.990398	0.104140	0.126903	0.006523	-0.008376	-0.035612	0.001408	0.010904	0.017356	0.000044
50	-2.205175	0.389416	0.989997	0.104129	0.124770	0.006333	0.000115	-0.026041	0.001007	0.010832	0.016230	0.000041
100	-2.211862	0.378772	0.990504	0.103484	0.128382	0.006538	-0.006572	-0.036685	0.001514	0.010741	0.017811	0.000045

Table 7: Simulation results for Weighted Log-normal Distribution $\mu = 3.958446,\ \sigma = 1.555012, c = 1.178781$ True values-Data Set-III

n	$\hat{\mu}$	$\hat{\sigma}$	$\hat{c}$	SE($\hat{\mu}$)	SE($\hat{\sigma}$)	SE($\hat{c}$)	Bias($\hat{\mu}$)	Bias($\hat{\sigma}$)	Bias($\hat{c}$)	MSE($\hat{\mu}$)	MSE($\hat{\sigma}$)	MSE($\hat{c}$)
20	3.908533	1.478545	1.169512	0.291574	0.274932	0.043394	-0.049913	-0.076467	-0.009269	0.087422	0.081359	0.001967
30	3.922165	1.512956	1.173642	0.231029	0.226182	0.034498	-0.036281	-0.042056	-0.005138	0.054638	0.052876	0.001215
40	3.937781	1.509222	1.172510	0.231104	0.212959	0.032605	-0.020665	-0.045790	-0.006271	0.053783	0.047403	0.001101
50	3.937113	1.518826	1.174977	0.223330	0.227427	0.035186	-0.021333	-0.036186	-0.003804	0.050282	0.052981	0.001251
100	3.948590	1.515137	1.172406	0.229000	0.216791	0.032663	-0.009856	-0.039875	-0.006375	0.052486	0.048542	0.001106

Criterion (HQIC), which are expressed as

$$AIC = 2k - 2\log L,$$

$$BIC = k \log n - 2\log L,$$

$$AICC = AIC + \frac{2k(k+1)}{n-k-1},$$

$$HQIC = -2\log L + 2k\log(\log(n)),$$

where k indicates the number of free parameters in the distribution, n is the sample size, and $(-\log L)$ refers to the maximized log-likelihood value of the fitted model. Table 9 reports the estimated parameters together with the values of the model selection criteria, offering a clear comparison among the competing distributions. Smaller values of AIC, BIC, AICC, HQIC, and $-2\log L$ suggest a superior model fit. In addition, Figure 3, 4, 5 displays the fitted probability density functions for Data Set I,II,III respectively.

5.1 Data Set-I:

The real dataset that we consider in this paper is taken from Ugarte et. al. [30]: 1501.82, 6989.43, 2424.02, 4150.29, 8693.35, 2643.77, 13148.37, 6149.39, 23587.21, 7248.37, 4788.22, 6009.51, 5349.65, 5741.32, 7065.81, 7261.37, 2358.42, 10357.88, 2499.05, 3022.90, 4234.86, 4482.03, 6363.71, 3329.91, 8740.47, 3664.95, 4515.97, 8497.71, 4569.89, 8069.63, 7366.79, 1525.41, 3363.02, 2420.57, 3576.74, 3708.05, 5819.12, 5479.38. The dataset represents estimates of leaf-mediated carbon storage, expressed in kilograms per hectare, collected from 38 sampling plots located in the mountainous zones of Navarra, Spain. Each plot is categorized according to forest composition: sites dominated almost entirely ($\geq$90%) by Fagus sylvatica are classified as monospecific stands, whereas those containing a diverse assemblage of tree species are designated as multispecific stands.

5.2 Data Set-II

The second dataset originates from the domain of reliability engineering and comprises the observed lifetimes of twenty mechanical components. These data have been examined in earlier studies, including the work of Murthy et al. [20]. The data are: 0.067, 0.068, 0.076, 0.081, 0.084, 0.085, 0.085, 0.086, 0.089, 0.098, 0.098, 0.114, 0.114, 0.115, 0.121, 0.125, 0.131, 0.149, 0.160 and 0.485.

5.3 Data Set-III

The third dataset considered in this study, originally reported by Linhart and Zucchini [16], comprises thirty recorded lifetimes associated with the operational failure of an aircraft's air-conditioning unit. These observations represent sequential failure durations and serve as a basis for evaluating the statistical properties of the proposed model. 23, 261, 87, 7, 120, 14, 62, 47, 225, 71, 246, 21, 42, 20, 5, 12, 120, 11, 3, 14, 71, 11, 14, 11, 16, 90, 1, 16, 52, 95.

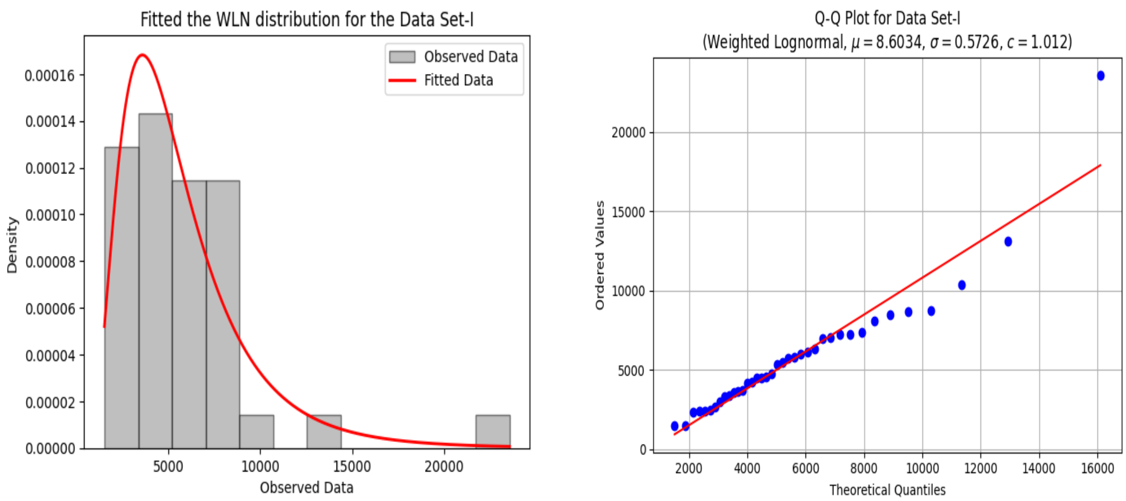


Figure 3: Fitted the weighted log-normal distribution and Q-Q Plot for the Data Set-I

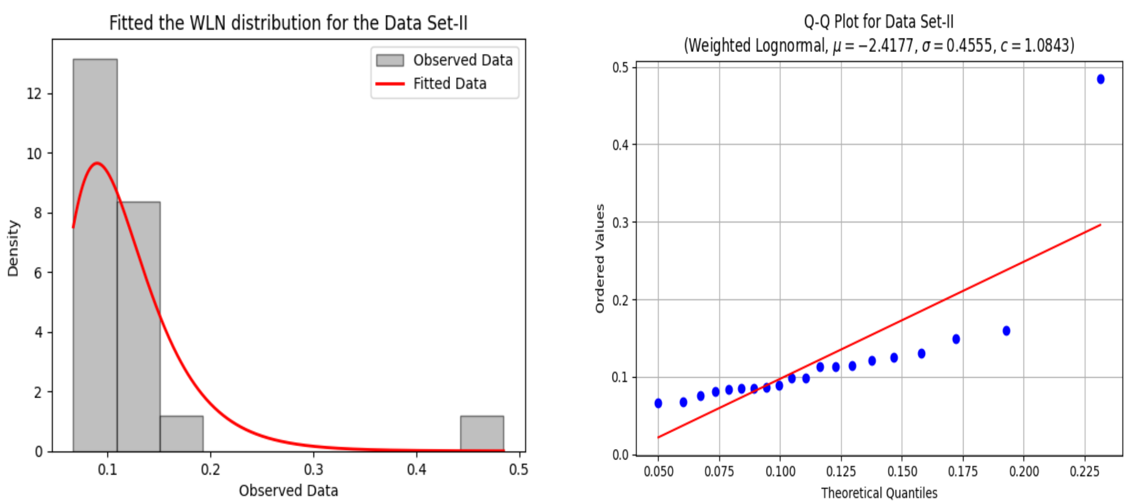


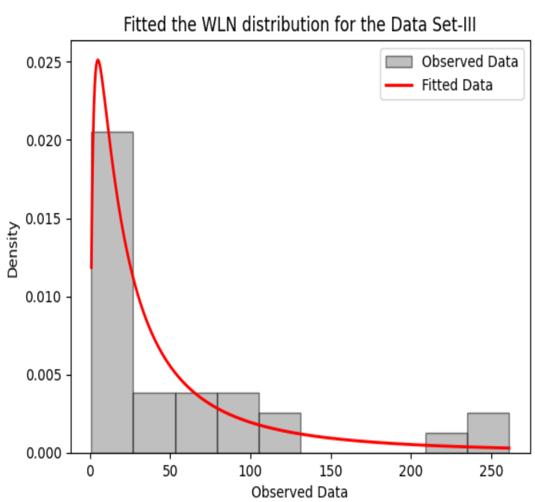
Figure 4: Fitted the weighted log-normal distribution and Q-Q Plot for the Data Set-II

Table 8: Summary of the data

Data	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	Skewness	Kurtosis
Data Set-I	1501.82	3416.45	5068.93	5808.38	7202.73	23587.21	2.6225	9.3646
Data Set-II	0.0670	0.0848	0.0980	0.1216	0.1220	0.4850	3.5855	12.2033
Data Set-III	1.00	11.25	20.50	58.93	83.00	261.00	1.6786	1.9197

Table 9: Estimated Parameters and Model Evaluation Criteria for WLN Distribution

Statistic	**Data Set-I**	**Data Set-II**	**Data Set-III**
μ (Location parameter)	8.6034	-2.4177	3.3741
σ (Scale parameter)	0.5726	0.4555	1.3255
c (weighted parameter)	1.0120	1.0843	1.0048
Log-Likelihood	-355.3246	33.5641	-151.6208
AIC	716.6493	-61.1282	309.2416
AICC	717.3552	-59.6282	310.1647
BIC	721.5620	-58.1410	313.4452
HQIC	718.3972	-60.5451	310.5864
KS Statistic	0.0760	0.1693	0.1048
KS P-value	0.9687	0.5584	0.8632



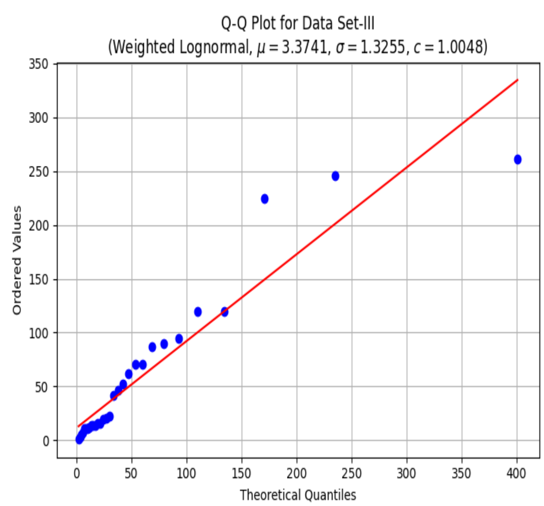


Figure 5: Fitted the weighted log-normal distribution and Q-Q Plot for the Data Set-III

6 Conclusions

As an extension of the standard log-normal family, this paper creates a three-parameter Weighted Log-Normal (WLN) distribution and determines its important analytical characteristics, such as its probability functions, survival structure, quantile form, and moment behavior. Maximum likelihood is used for parameter estimation for optimization to ensure numerical accuracy. Applications to mechanical failure data and environmental measurements show the model's good empirical performance, while simulation trials confirm the estimators' stability and dependability. The WLN distribution offers a considerably better fit across all datasets analyzed, demonstrating its efficacy as a workable and adaptable substitute for modeling mechanical and environmental events.

Declarations

Conflicts of Interest: The authors declare no competing interests.

Consent for Publication: I consent to the publication of the data and materials.

Data Availability This study did not generate or utilize any new datasets; therefore, data sharing is not applicable.

Code Availability We are using the software Python to calculate the numerical values and graphics.

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